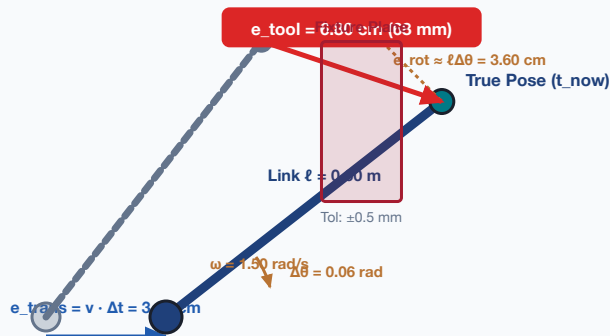


COORDINATE TRANSFORM STALENESS & LEVER-ARM ERROR PROPAGATION

Rigid-Body Kinematic Divergence: $e_{\text{tool}} \approx v \cdot \Delta t + \ell \cdot \omega \cdot \Delta t$ Under Stale Coordinate Frames

PHYSICAL ARM DYNAMICS & LEVER ARM DIVERGENCE



CRITICAL COLLISION: 68.0 mm Error vs 0.5 mm Mechanical Clearance (136x Breach)

Uncompensated 40 ms lag drives end-effector into solid fixture during trajectory execution.

KINEMATIC LEVER-ARM ARITHMETIC

• Translational Drift:

$$e_{\text{trans}} = v \cdot \Delta t = 0.80 \text{ m/s} \times 0.040 \text{ s} = 3.20 \text{ cm}$$

• Rotational Arc Deflection:

$$e_{\text{rot}} \approx \ell \cdot (\omega \cdot \Delta t) = 0.60 \text{ m} \times (1.50 \times 0.040) = 3.60 \text{ cm}$$

• Total Tool-Point Offset:

$$e_{\text{tool}} \approx e_{\text{trans}} + e_{\text{rot}} = 3.20 + 3.60 = 6.80 \text{ cm} \text{ (68.0 mm)}$$

• Task Safety Clearance:

$$e_{\text{tol}} = 0.50 \text{ mm} \text{ (Exceeded by 136x within 40 ms!)}$$

LOUD VS SILENT TRANSFORM FAILURES

✓ LOUD FAILURE: MISSING / DROPPED TRANSFORM

- Sensor link drops; frame tree records null edge.
- Matrix traversal raises `FRAME_NOT_FOUND` exception.
- Result: Deterministic emergency stop / safe position hold.

✗ SILENT FAILURE: FROZEN / REPUBLISHED MATRIX

- Middleware republishes cached pose with fresh OS timestamp.
- Downstream linear algebra evaluates with zero exceptions.
- Result: Trajectory planner drives tool into solid fixture.

RULE: Transforms **MUST** carry immutable evidence epoch `t_e`.